

GGPlot For Visualization

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Background

In this notebook, we™ review a scenario, and use ggplot2 to quickly create data visualizations that allow us to explore our data and gain new insights. We will use basic ggplot2 syntax and perform data visualization in R.

The Scenario

In this scenario, we are junior data analysts working for a hotel booking company. we have cleaned and manipulated our data, and gotten some initial insights we would like to share. Now, we are going to create some simple data visualizations with the ggplot2 package. we will use basic ggplot2 syntax and troubleshoot some common errors we might encounter.

Step 1: Import our data

The data in this example is originally from the article Hotel Booking Demand Datasets

(<https://www.sciencedirect.com/science/article/pii/S2352340918315191>

(<https://www.sciencedirect.com/science/article/pii/S2352340918315191>)), written by Nuno Antonio, Ana Almeida, and Luis Nunes for Data in Brief, Volume 22, February 2019.

The data was downloaded and cleaned by Thomas Mock and Antoine Bichat for #TidyTuesday during the week of February 11th, 2020 (<https://github.com/rfordatascience/tidytuesday/blob/master/data/2020/2020-02-11/readme.md> (<https://github.com/rfordatascience/tidytuesday/blob/master/data/2020/2020-02-11/readme.md>)).

we can learn more about the dataset here: <https://www.kaggle.com/jessemostipak/hotel-booking-demand> (<https://www.kaggle.com/jessemostipak/hotel-booking-demand>)

Import data from a .csv in the project folder called “hotel_bookings.csv” and save it as a data frame called hotel_bookings :

```
hotel_bookings <- read.csv("hotel_bookings.csv")
```

Step 2: Look at a sample of our data

Use the head() function to preview our data:

```
head(hotel_bookings)
```

```

##      hotel is_canceled lead_time arrival_date_year arrival_date_month
## 1 Resort Hotel      0      342      2015      July
## 2 Resort Hotel      0      737      2015      July
## 3 Resort Hotel      0       7      2015      July
## 4 Resort Hotel      0      13      2015      July
## 5 Resort Hotel      0      14      2015      July
## 6 Resort Hotel      0      14      2015      July
## arrival_date_week_number arrival_date_day_of_month stays_in_weekend_nights
## 1      27      1      0
## 2      27      1      0
## 3      27      1      0
## 4      27      1      0
## 5      27      1      0
## 6      27      1      0
## stays_in_week_nights adults children babies meal country market_segment
## 1      0      2      0      0 BB PRT Direct
## 2      0      2      0      0 BB PRT Direct
## 3      1      1      0      0 BB GBR Direct
## 4      1      1      0      0 BB GBR Corporate
## 5      2      2      0      0 BB GBR Online TA
## 6      2      2      0      0 BB GBR Online TA
## distribution_channel is_repeated_guest previous_cancellations
## 1      Direct      0      0
## 2      Direct      0      0
## 3      Direct      0      0
## 4      Corporate      0      0
## 5      TA/T0      0      0
## 6      TA/T0      0      0
## previous_bookings_not_canceled reserved_room_type assigned_room_type
## 1      0      C      C
## 2      0      C      C
## 3      0      A      C
## 4      0      A      A
## 5      0      A      A
## 6      0      A      A
## booking_changes deposit_type agent company days_in_waiting_list customer_type
## 1      3 No Deposit NULL NULL 0 Transient
## 2      4 No Deposit NULL NULL 0 Transient
## 3      0 No Deposit NULL NULL 0 Transient
## 4      0 No Deposit 304 NULL 0 Transient
## 5      0 No Deposit 240 NULL 0 Transient
## 6      0 No Deposit 240 NULL 0 Transient
## adr required_car_parking_spaces total_of_special_requests reservation_status
## 1 0 0 0 Check-Out
## 2 0 0 0 Check-Out
## 3 75 0 0 Check-Out
## 4 75 0 0 Check-Out
## 5 98 0 1 Check-Out
## 6 98 0 1 Check-Out
## reservation_status_date
## 1 2015-07-01
## 2 2015-07-01

```

```
## 3      2015-07-02
## 4      2015-07-02
## 5      2015-07-03
## 6      2015-07-03
```

We can also use `colnames()` to get the names of all the columns in our data set.

```
colnames(hotel_bookings)
```

```
## [1] "hotel" "is_canceled"
## [3] "lead_time" "arrival_date_year"
## [5] "arrival_date_month" "arrival_date_week_number"
## [7] "arrival_date_day_of_month" "stays_in_weekend_nights"
## [9] "stays_in_week_nights" "adults"
## [11] "children" "babies"
## [13] "meal" "country"
## [15] "market_segment" "distribution_channel"
## [17] "is_repeated_guest" "previous_cancellations"
## [19] "previous_bookings_not_canceled" "reserved_room_type"
## [21] "assigned_room_type" "booking_changes"
## [23] "deposit_type" "agent"
## [25] "company" "days_in_waiting_list"
## [27] "customer_type" "adr"
## [29] "required_car_parking_spaces" "total_of_special_requests"
## [31] "reservation_status" "reservation_status_date"
```

Step 3: Install and load the ‘ggplot2’ package

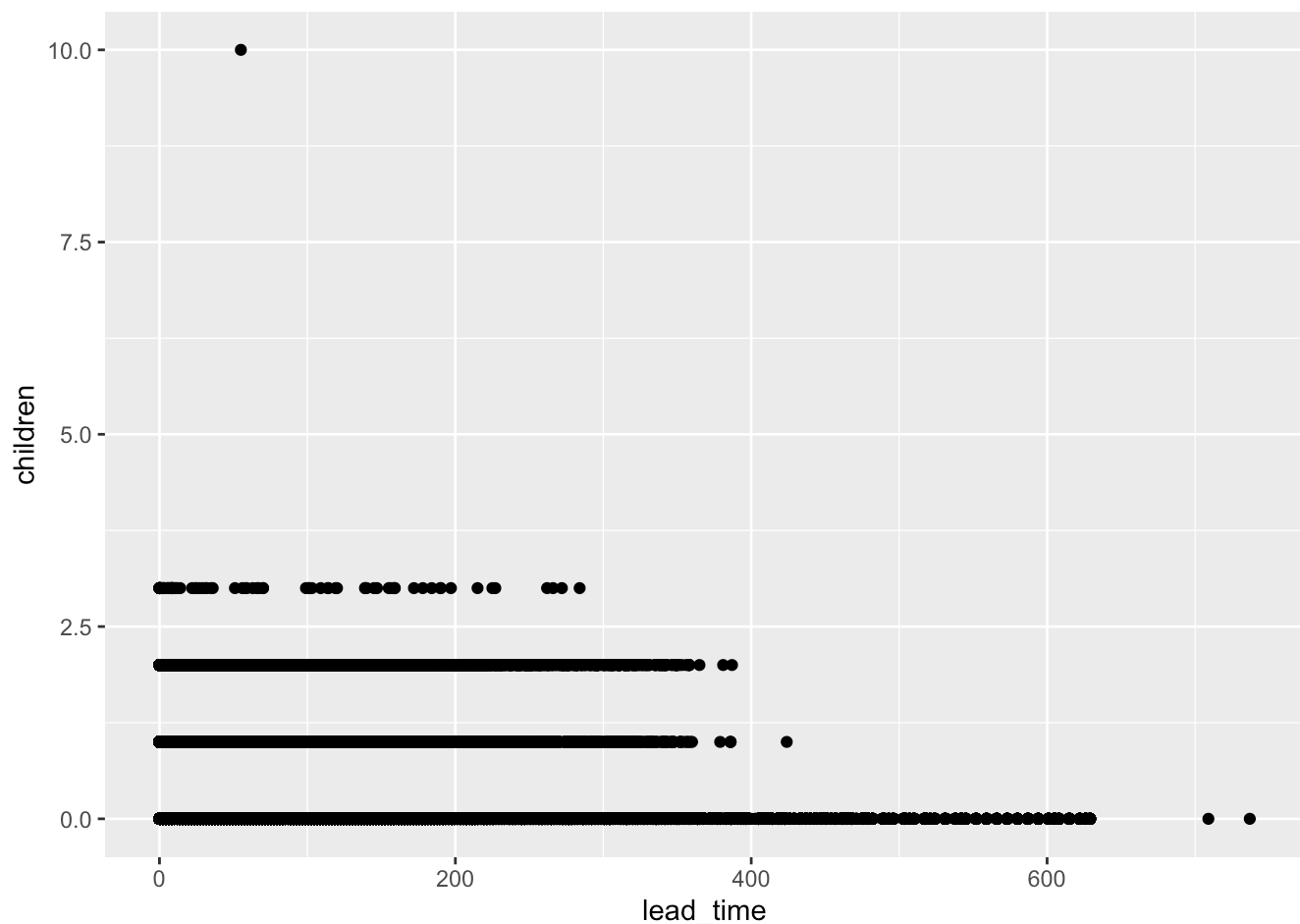
If we haven’t already installed and loaded the `ggplot2` package, we will need to do that before we can use the `ggplot()` function.

Step 4: Begin creating a plot

We can use `ggplot2` to determine if people with children book hotel rooms in advance.

```
ggplot(data = hotel_bookings) +
  geom_point(mapping = aes(x = lead_time, y = children))
```

```
## Warning: Removed 4 rows containing missing values (`geom_point()`).
```



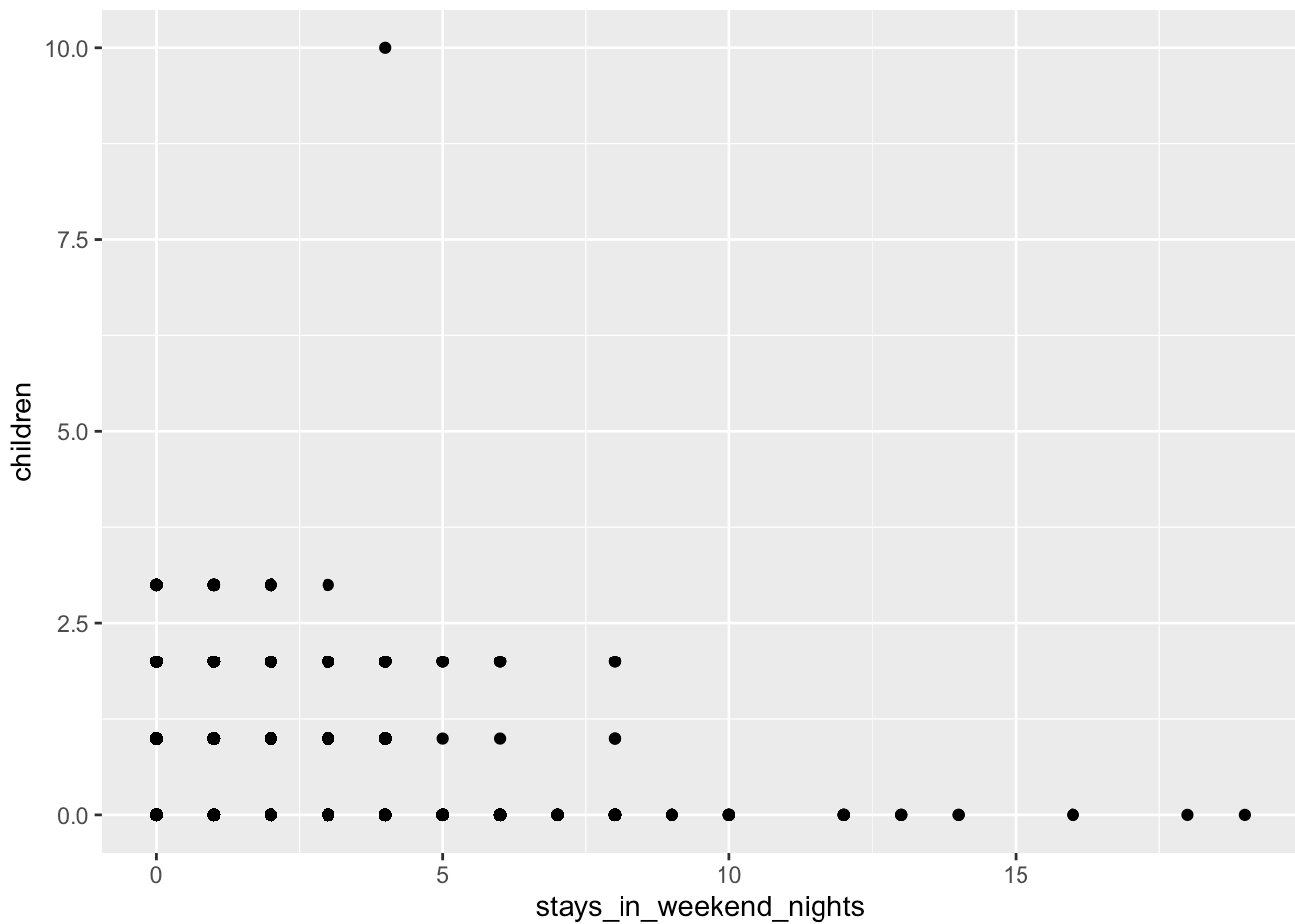
On the x-axis, the plot shows how far in advance a booking is made, with the bookings furthest to the right happening the most in advance. On the y-axis it shows how many children there are in a party.

Step 5: Try mapping 'stays_in_weekend_nights' on the x-axis

Try mapping 'stays_in_weekend_nights' on the x-axis and 'children' on the y-axis

```
ggplot(data = hotel_bookings) +
  geom_point(mapping = aes(x = stays_in_weekend_nights, y = children))
```

```
## Warning: Removed 4 rows containing missing values (`geom_point()`).
```



We now have a scatterplot with 'stays_in_weekend_nights' on the x-axis and 'children' on the y-axis.

Activity Wrap Up

The `ggplot2` package allows us to quickly create data visualizations that can answer questions and give us insights about our data.